Week 9. By Deng Yufan.

Prob 1. Prove that TQBF restricted to formulas where following the quantifier is CNF is PSPACE-Complete.

Obviously the restricted version (called TQBF') $\leq_p$ TQBF. We prove TQBF $\leq_p$ TQBF'.

Consider the formular φ in the end of TQBF. We write its expression tree.

For every arithmetic node, we create a $\exists$ quantifier in the very end.

(this can be done by inserting never-used virable in $\forall$).

then we write $y_1 \Leftrightarrow (x_1 \oplus x_2)$ as $(y_1 \wedge L_1) \vee (y_1 \wedge L_2) \cdots \vee (\neg y_1 \wedge L_1') \vee \cdots$.
where $L_1 \vee L_2 \vee \cdots$ is the CNF for $x_1 \oplus x_2$. $L_1' \vee L_2' \vee \cdots$ for $\neg(x_1 \oplus x_2)$.
the length of the CNF is a constant, thus the CNF for $y_1 \Leftrightarrow (x_1 \oplus x_2)$ is of constant length.

thus every node can be writen as $y_k \Leftrightarrow (y_i \oplus y_j)$. and we only need Poly space to memorize the new CNF.

Prob 2. $SUM = \{\langle x, y, z\rangle \mid x, y, z > 0,\ x+y=z\}$. Prove $SUM \in L$.

We give a algo. to solve SUM.

for input x, y, z. memorize the current place from low bit to high bit.

and whether there is a carry.

calc $x_i + y_i + carry \rightarrow z_i + carry'$. if z_i is wrong, reject. otherwise forget carry and memorize carry'.

If all digit are verified, accept.

Prob 3. a. Prove a graph is bipartite iff no odd ring.

only if: If there is a odd ring, then we can start at vertice x. go odd steps and back to x. If x is at left, we are at right after odd steps. but we go back to x. contradiction.

if: suppose the graph is connected. take a vertice x to be root, and pick any of spanning tree. if a vertice has odd depth, then assigned to left. even. right. then every vertex is assigned. if odd depth x and odd depth y is connected, then there is a odd ring $x \xrightarrow{\text{edge}} y \xrightarrow{\text{tree (even length)}} x$. likewise, even.

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* 6. BIPARTITE = $\{\langle G\rangle \mid G$ is bipartite$\}$. Prove BIPARTITE $\in$ NL.

Consider for all $G=(V,E)$, construct $G' = (V \times \{0,1\}, E')$.

that $E' = \{((x,0),(y,1)) \mid (x,y) \in E\}$. (notice that E is undirected. thus $(x,y)\in E \Rightarrow (y,x)\in E$).

This reduction can be done in L obviously. then judge if $\exists x$, $(x,0)$ can go to $(x,1)$. If (iff $\exists$ odd ring) exists, then reject. otherwise accept. (since G' is undirected. this can be done in L not NL).

So BIPARTITE is NL.

Prob 4. Let $S(n) \geq \log n$ be a space-constructible func. Prove

$NL = coNL \Rightarrow NSPACE(S(n)) = coNSPACE(S(n))$.

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6. BIPARTITE = $\{\langle G\rangle \mid G$ is bipartite$\}$. Prove it in NL.

Consider $\overline{\text{BIPARTITE}}$. We construct $G' = (V \times \{0,1\}, E')$ where

$E = \{((x,0),(y,1)) \mid (x,y) \in E\}$. and guess a vertex x. if $(x,0)$ is connected with $(x,1)$ (iff exist odd ring), then accept. Otherwise reject.

Thus $\overline{\text{BIPARTITE}} \in NL$. since $NL = coNL$, BIPARTITE $\in$ NL.

Prob 4. Let $S(n) \geq \log n$ be space-constructible. Prove $NL = coNL \Rightarrow NSPACE(S(n)) = coNSPACE(S(n))$.

We prove $P \in NL$ iff $\bar{P} \in NL \Rightarrow P \in NSPACE(S(n))$ iff $P \in coNSPACE(S(n))$.

For all $P \in NSPACE(S(n))$ [$P \in coNSPACE(S(n))$], there exists $G=(V,E)$ that $|V| = 2^{O(S(n))}$, E is directed and denotes transfer between IDs.

To solve P, we need to determine whether S can reach for a accept vertex [reject vertex].

For a graph G. determine S reach for any of $x \in F$ is NL. so determine S cannot reach any $x \in F$ is $coNL = NL$.

To solve $\bar{P}$, we only need to determine whether S cannot reach any accept vertex [reject vertex]. this can be done by NTM in space $O(\log(2^{O(S(n))})) = O(S(n))$.

So $\bar{P} \in NSPACE(S(n))$ [$\bar{P} \in coNSPACE(S(n))$].

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